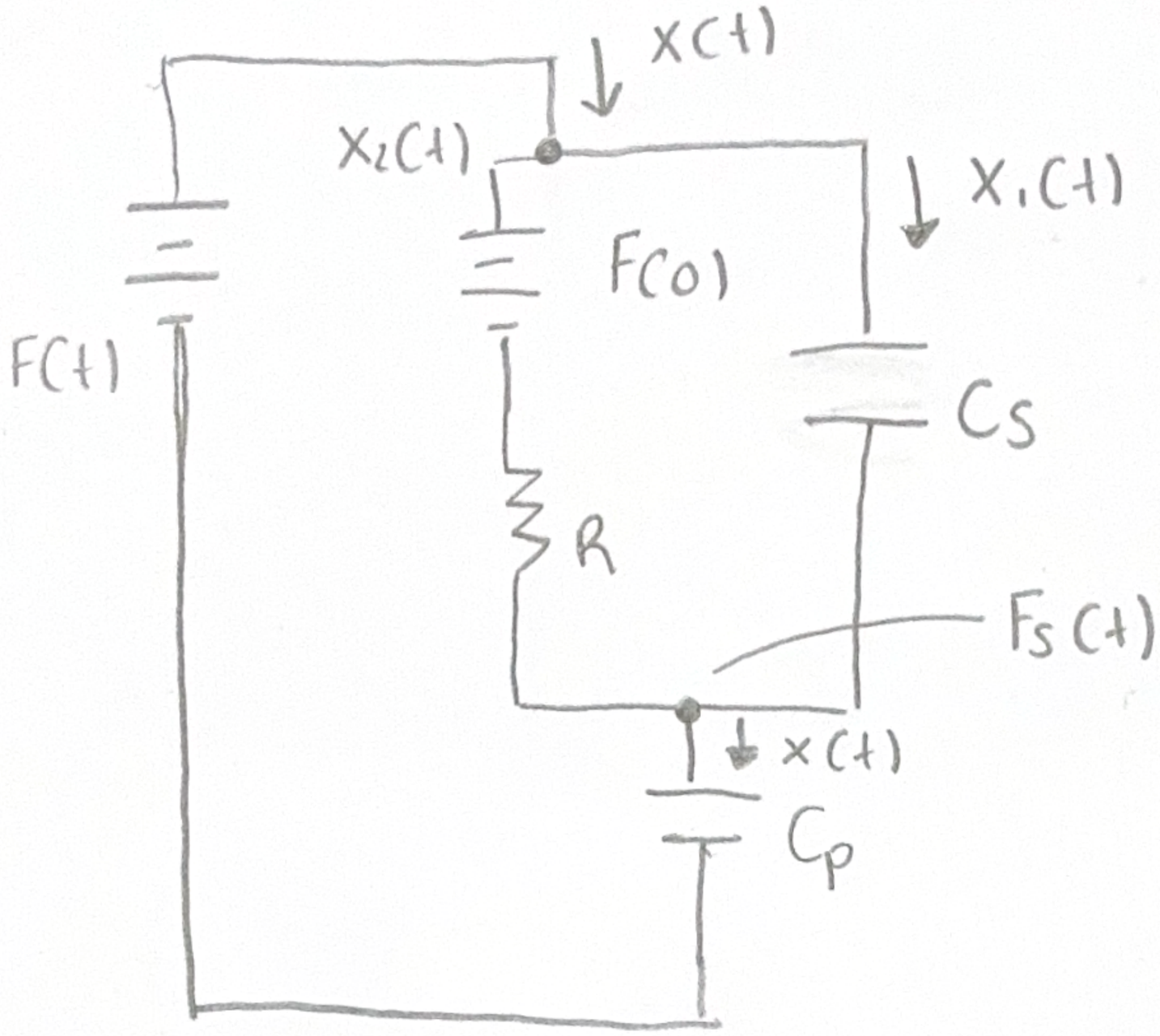
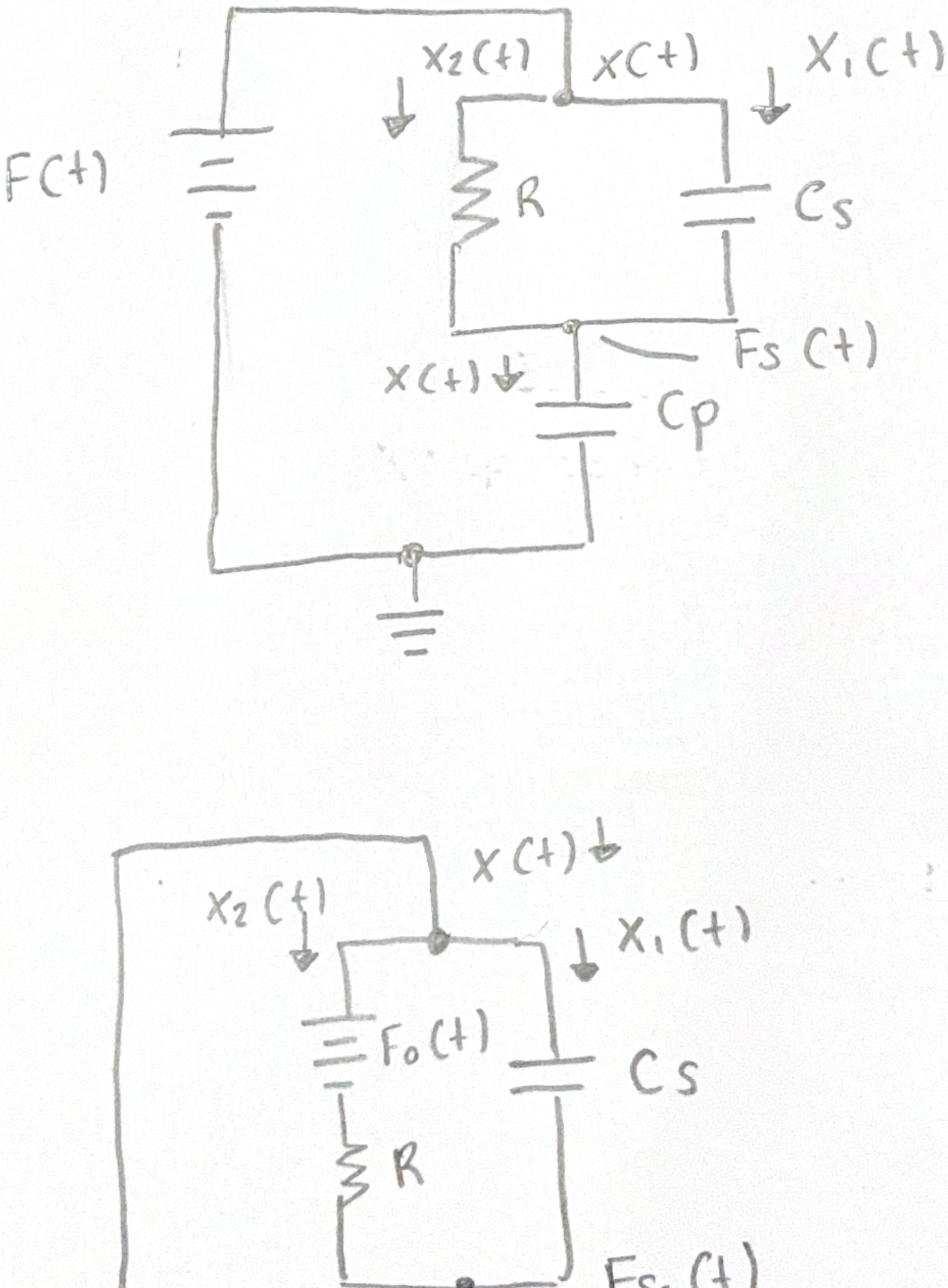
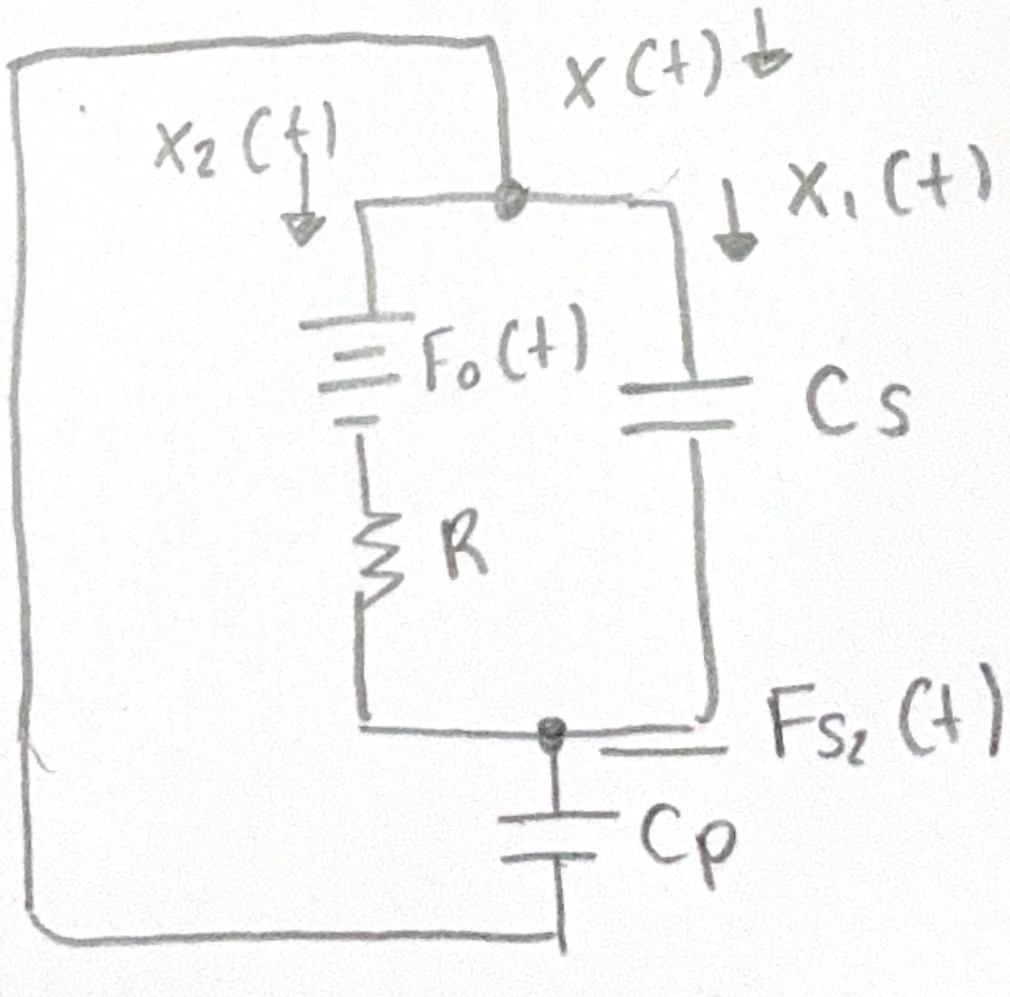
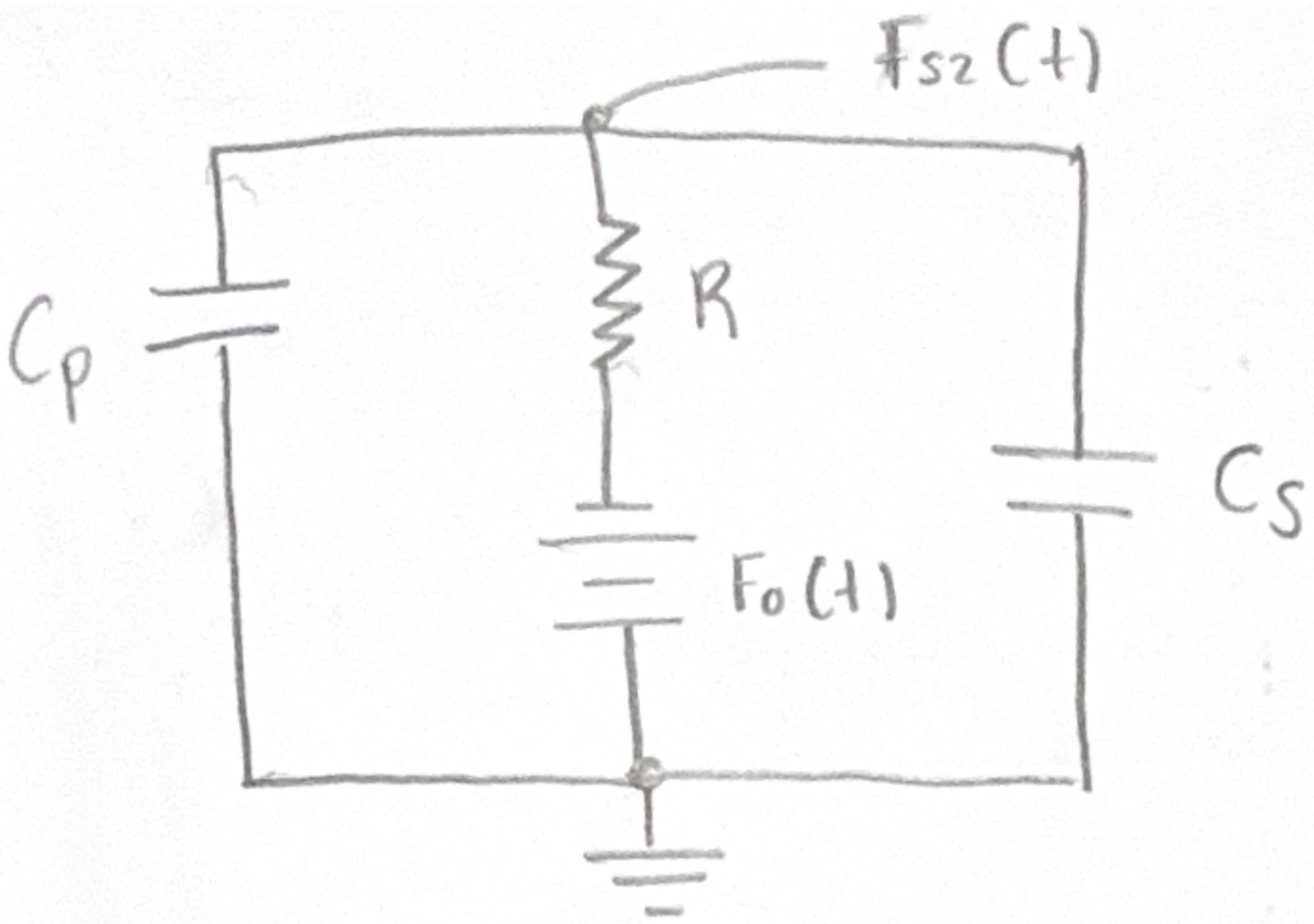
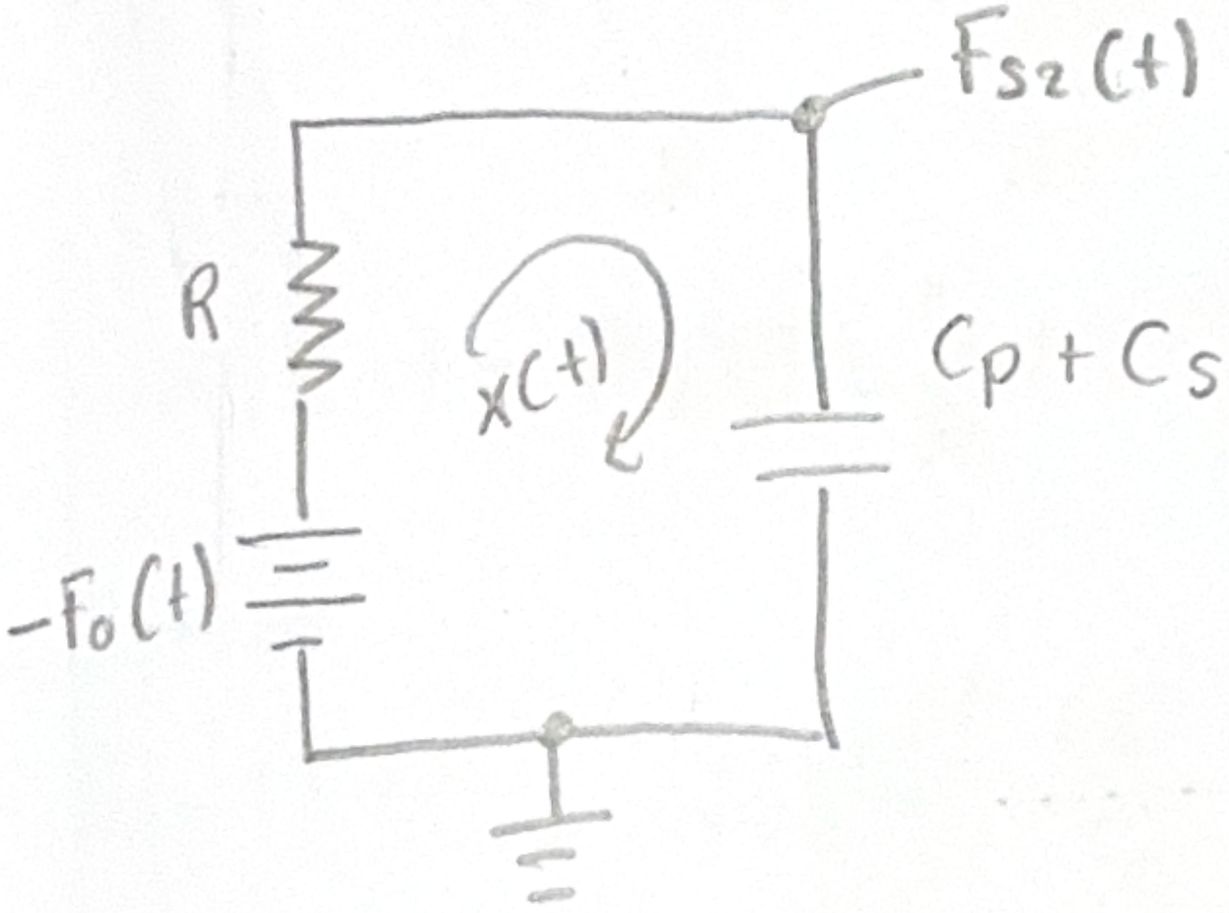
$x(t) = X_1(t) + X_2(t)$ analisis circuito











analisis matemático

debido al presentarse dos fuentes de voltaje, aplicamos el principio de superposición.

Circuito 1. Apagando $F_o(t)$

$$x(t) = X_1(t) + X_2(t)$$

$$x(t) = C_p \frac{dF_{s_1}(t)}{dt}$$

$$X_1(t) = C_s \frac{d[F(t) - F_{s_1}(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s_1}(t)}{R}$$

$$C_p \frac{d F_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_{ps} F_{s1} = C_{ss}[F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_{ps} F_{s1} = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$\left(C_{ps} + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_{ss} + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R}}{\frac{R(C_p + C_s)s + 1}{R}} F(s)$$

$$F_{s1}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Circuito 2: Apagando $F(t)$

recordando que $F_o(t) = \alpha F(t)$

$$-\alpha F(t) = Rx(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s_2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$-\alpha F(s) = Rx(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{1}{\alpha}\left(R + \frac{1}{(C_p + C_s)s}\right) x(s) = -\frac{1}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\, x(s)$$

$$\frac{F_{s_2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{1}{\alpha}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\, x(s)} = -\frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

Función de transferencia

$$F_s(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R_s + 1}{R(C_p + C_s)s + 1} F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R_s + 1 - \alpha}{R(C_p + C_s)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} s F(s) \left(1 - \frac{F_s(s)}{F(s)}\right)$$

$$= \lim_{s \to 0} s \frac{1}{s} \left(\frac{1 - RC_s s + 1 - \alpha}{(RC_p + RC_s)s + 1}\right) = \alpha$$

Estabilidad en lazo abierto

$$(RC_p + RC_s)s + 1 = 0$$

$$s = -\frac{1}{RC_p + RC_s}$$

El sistema es estable.